Energy Consumption Time Series Forecasting

Hourly Demand Prediction with Lag Features and Gradient Boosting

Felipe Taha Sant'Ana

Agenda

1. Business Context
2. Dataset & EDA
3. Feature Engineering
4. Results
5. Conclusions

Why Forecasting Matters

The Stakes

- Under-prediction $\rightarrow$ **blackouts** and grid instability
- Over-prediction $\rightarrow$ **wasted resources** and higher costs
- Accurate forecasts enable optimal **dispatch scheduling**

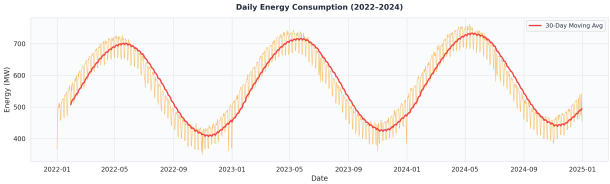
Target Metrics

KPI	Target
MAPE	$< 5\%$
R^2	> 0.75
Horizon	1 hour ahead
Latency	< 100ms

Our Goal

Build a model that forecasts hourly energy demand with **less than 5% MAPE**, using weather data, calendar features, and lag-based time series features.

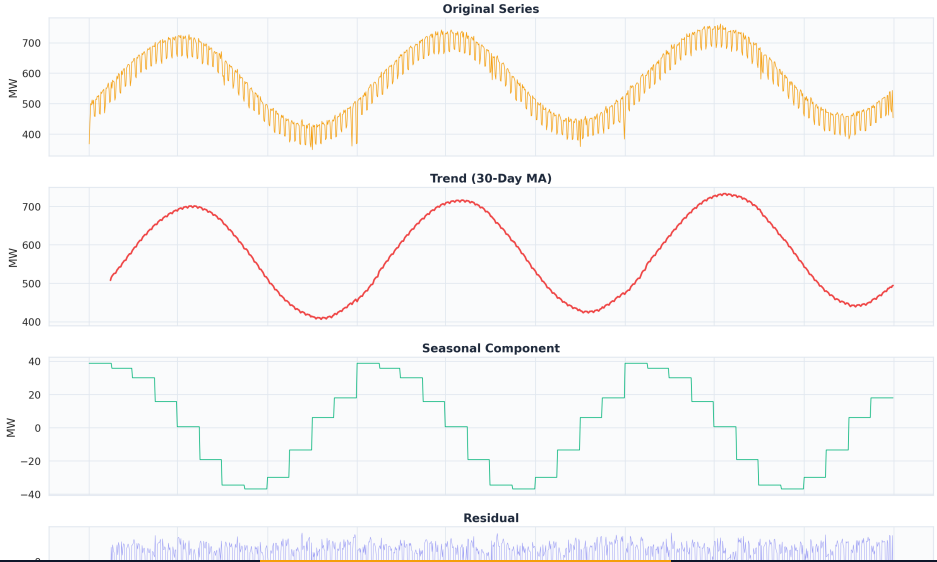
3 Years of Hourly Data



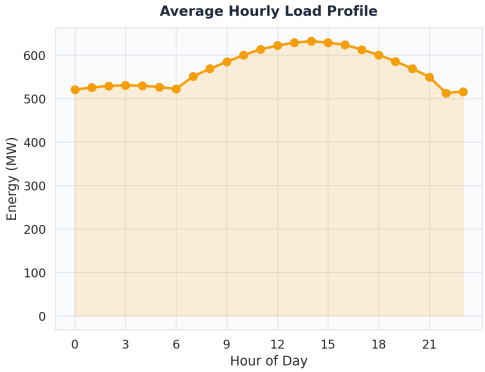
Dataset

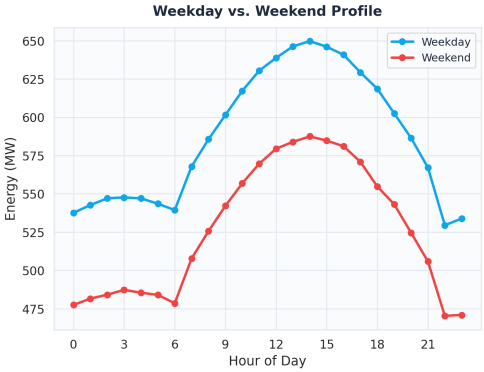
- **26,304** hourly observations
- **2022–2024** (3 full years)
- Weather: temperature, humidity, wind, cloud cover
- Calendar: hour, day-of-week, month, weekend flag
- Energy range: 236–891 MW

Seasonal Decomposition



Daily Load Profiles

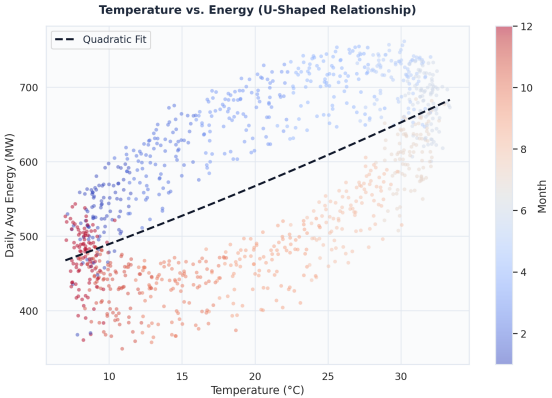


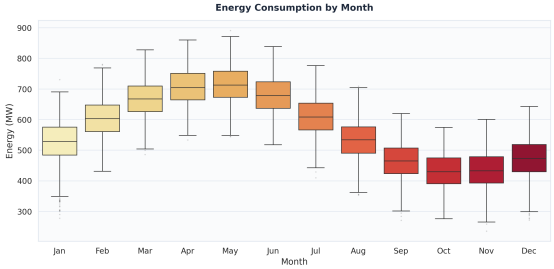


Key Patterns

Peak demand occurs at **mid-afternoon (14:00–16:00)**. Weekday loads are **significantly higher** than weekends. The daily cycle is the dominant short-term pattern.

Temperature Relationship & Monthly Variation





U-Shaped Temperature Effect

Energy demand increases at **both extremes**: cold weather drives heating load, hot weather drives cooling load. This non-linear relationship is captured via a quadratic temperature term.

Feature Engineering Strategy

Cyclical Encodings

- Hour $\rightarrow$ sin/cos (24h cycle)
- Month $\rightarrow$ sin/cos (12mo cycle)
- Day-of-week $\rightarrow$ sin/cos (7d cycle)
- Temperature^2 for U-shape

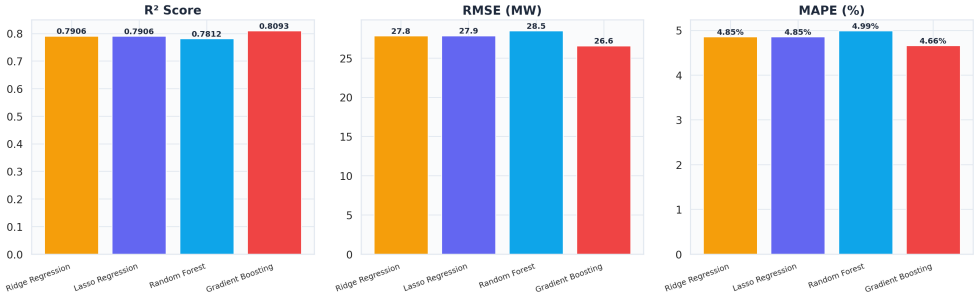
Lag Features

- `Lag_1h` — previous hour
- `Lag_24h` — same hour yesterday
- `Lag_168h` — same hour last week
- `RollMean_24h` — 24h rolling avg
- `RollStd_24h` — 24h rolling std

Train/Test Split

Time-based split: **Jan 2022 – Sep 2024** for training, **Oct – Dec 2024** for testing (2,208 hours). No data leakage guaranteed.

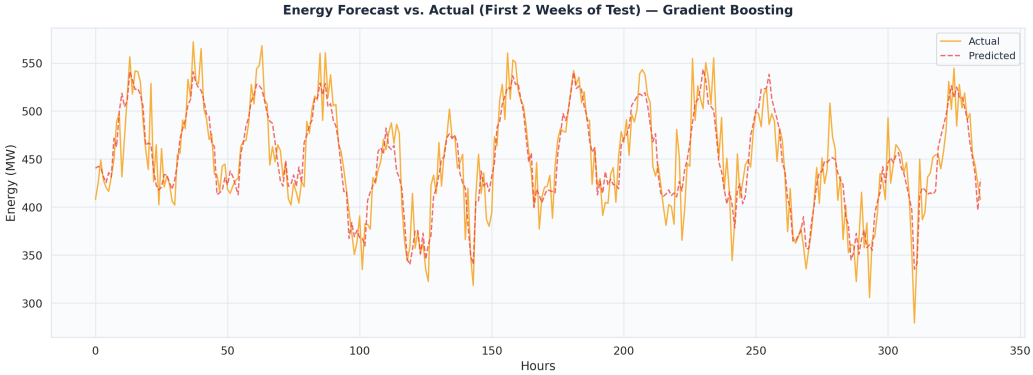
Model Comparison



Winner: Gradient Boosting

$R^2 = 0.809$ RMSE = 26.6 MW MAE = 19.7 MW MAPE = 4.66%

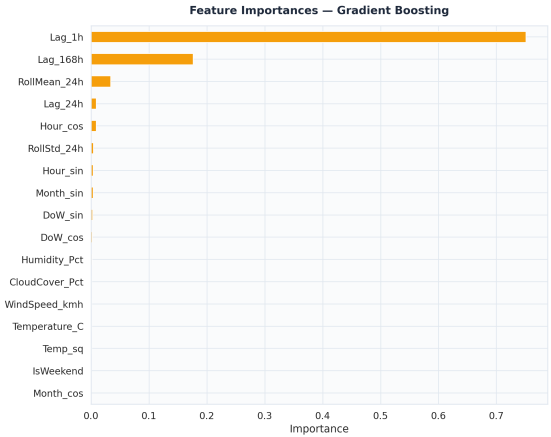
Forecast vs. Actual

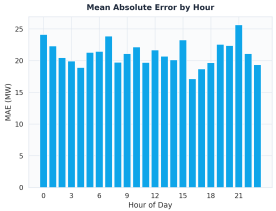


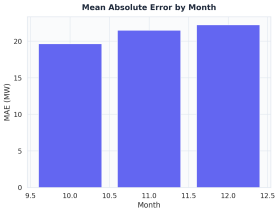
Analysis

The model closely tracks the actual demand curve, capturing daily peaks and troughs. Largest errors occur during rapid demand transitions (morning ramp-up, evening ramp-down).

Feature Importance & Error Analysis







Key Takeaways

Lag features dominate: the 1-hour lag is the single strongest predictor. Errors are highest

Summary & Future Work

Results

- MAPE of **4.66%** (target: $<5\%$)
- Lag features + cyclical encoding = key
- Temperature^2 captures non-linearity
- Time-based split prevents leakage

Future Roadmap

1. LSTM / Temporal Fusion Transformer
2. Probabilistic forecasts (quantile reg.)
3. Multi-step horizons (24h, 7d)
4. External data: holidays, solar/wind
5. Online learning with streaming

Thank You

Questions & Discussion

GitHub: github.com/ftahas
Email: ftahas@proton.me
LinkedIn: linkedin.com/in/ftsantana